Marlon Esteban Brenes Rojas

ACADEMIC DOSSIER & RESEARCH PUBLICATION BRIEF

This dossier serves as formal documentation of peer-reviewed academic accomplishments, curated for review by university hiring committees, graduate admissions panels, and prospective faculty collaborators. It establishes a proven track record of high-impact research at the intersection of Artificial Intelligence, Software Engineering, and Practical DevSecOps.

Paper Title:	**Automating Software Supply Chain Security: AI-Powered Threat Detection in DevSecOps**
Authors:	Marlon Brenes Rojas, Sara [Co-Author]
Venue:	ACM International Secure Development Conference (SecDev 2026)
Track:	Technical Papers Track (Peer-Reviewed, 100% Open Access publishing model via ACM Digital Library)
Official Link:	**conf.researchr.org/track/secdev-2026/secdev-2026-papers**
Timeline/Location:	July 5 – July 6, 2026 | Concordia SGW Campus, Montreal, Canada (Co-located with FSE 2026)

Executive Research Abstract

Modern enterprise software distribution chains are increasingly threatened by decentralized, multi-tiered dependency graphs that introduce significant attack surfaces. Traditional static analysis tools and classic break-and-patch paradigms are demonstrably inadequate at keeping pace with dynamic runtime risks, sophisticated malicious dependency injections, and sudden software supply chain integrity failures.

This paper presents a novel framework that bridges academic formal verification with industrial deployment workflows. We introduce a mechanism for embedding localized Small Language Models (SLMs) and automated reasoning systems directly into continuous deployment pipelines as non-bypassable security guardrails. By engineering automated, AI-driven behavioral threat detection capabilities natively into runtime orchestration architectures, this framework successfully charts dependency anomalies in real time and strictly enforces programmatic Policy-as-Code to prevent regression vectors prior to compilation.

Core Technical Contributions

- **Constructive Security Architecture:** Aligns directly with the core paradigm shift of building security *into* systems from inception, migrating industry standards away from purely reactive post-incident scanning.
- **Localized Agent Telemetry:** Outlines innovative implementations utilizing local language models to observe and evaluate deep behavior patterns across dependency boundaries safely, minimizing third-party data leakage risks.
- **Continuous Orchestration Integration:** Offers a structured, scalable blueprint showing how automated reasoning engines interface directly with complex infrastructure automation frameworks and modern enterprise pipelines.

Academic and Institutional Relevance

This research provides concrete evidence of ability to conceptualize, author, and defend peer-reviewed computing frameworks matching global cybersecurity mandates. It addresses critical public sector priorities and national initiatives regarding closing software understanding gaps through automated verification and scalable proof patterns. The architectural concepts established within this paper open significant potential for federally funded university research grants and corporate-academic security consortium sponsorships.

Contextual Note for Reviewing Academic Committees: *The ACM Secure Development Conference (SecDev) is highly selective and uniquely emphasizes the development of secure computing systems. Being co-located with FSE 2026 places this research immediately alongside leading global institutional contributions. Verification of track acceptances and peer review outcomes can be cross-audited via the official conference repository listing.*